

Original Healthcare Dataset

Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number
jeFFReY WoOd	81	Female	AB-	Cancer	07 July 2021	Amber Holmes	Brown-Yoder	UnitedHealthcare	41982.3743962326	
luURie EdWaRdS	51	Female	A-	Hypertension	08 October 2022	Kevin Mack	Armstrong Everett, Adams and	UnitedHealthcare	24786.218020918	
gEOReG PHILIPS	69	Female	A-	Asthma	21 March 2021	Timothy Guerra	Collins-Gutierrez	UnitedHealthcare	11202.3466921343	
tanMY aLUaREZ	80	Female	A+	Diabetes	08 August 2023	Scott Porter	Ltd Brown	UnitedHealthcare	49446.5648075073	
juLIe HarVey	18	Female	B-	Asthma	06 May 2021	Tammy Wilson	Walton Inc	UnitedHealthcare	40391.3345061197	
DaVid Levine	59	Female	AB+	Asthma	30 March 2023	Mary Brown	Williams-Garcia	UnitedHealthcare	40998.7554024913	
ROBIN MannIng	56	Female	AB+	Hypertension	31 May 2023	Yvonne Harper	Allen Campbell, and Hurst	UnitedHealthcare	1583.37755689399	
BRADLeY buRKE	37	Female	B+	Cancer	29 August 2023	Alvin Young	Inc Clark	UnitedHealthcare	40195.471981516	
RICHarD RODRIGUEZ	49	Female	AB-	Diabetes	07 September 2021	Charles Austin	Inc Wright	UnitedHealthcare	8459.08764254116	
coRey orIega	27	Female	B-	Arthritis	24 April 2023	Adam Whitehead	Parker-Coleman	UnitedHealthcare	19017.645580705	
BenYf vASQUEZ	44	Female	O+	Hypertension	22 November 2020	Donna Simpson	Group Martin	UnitedHealthcare	17964.5695428175	
maRY marTineZ	57	Female	O+	Arthritis	15 April 2021	Chad Kim	Ramirez PLC	UnitedHealthcare	34892.8058494266	
ILUCAS mooRE	74	Female	O-	Hypertension	03 January 2023	Alexander Sanchez	Reid-Lara	UnitedHealthcare	11987.1936460421	
TrAd WILLIAMS	32	Female	A-	Diabetes	26 February 2021	Ashley Osborn	Powers Webb Blackwell, and	UnitedHealthcare	32444.3149971729	
seth SchmiDt	30	Female	B+	Hypertension	14 January 2023	Julie Herrera	Ltd Evans	UnitedHealthcare	46154.1398165559	
conNoR VincEnt	33	Female	A+	Arthritis	02 March 2024	Devin Williams	and Shaffer Williams Leon,	UnitedHealthcare	43558.7670527172	
aaron oMingUeZ	70	Female	B-	Hypertension	14 November 2021	Matthew West	Kim and Sons	UnitedHealthcare	29977.9124464471	
TERRY kLine	39	Female	A+	Arthritis	11 April 2020	Richard Avery	Bowman-Molina	UnitedHealthcare	48522.9291457135	
bRIAN bENNeTt	43	Female	AB+	Cancer	04 May 2022	Keith Young	Harrison-Gardner	UnitedHealthcare	19928.5504808258	
nataSHA SMITH	18	Female	A-	Diabetes	23 July 2020	Benjamin Perez	Smith Harvey Wells, and	UnitedHealthcare	23840.3109166424	
MIChElE WILLIAMS	75	Female	A-	Hypertension	26 December 2020	Anthony Aguilar	Stephens-Ramos	UnitedHealthcare	10970.6234171714	
brANDOn DALton	36	Female	O-	Cancer	25 September 2020	Brett Stokes	Rogers-Davis	UnitedHealthcare	21216.4026845777	
HOLBY kniGHt	47	Female	B+	Arthritis	04 March 2022	Paul Stevenson	Inc Everett	UnitedHealthcare	3444.63514994028	
seAN waRe	45	Female	A-	Hypertension	15 June 2019	Timothy Miller	Matthews-Foster	UnitedHealthcare	30952.858792672	
JENNIfeR BoYd	25	Female	O-	Obesity	05 June 2021	Brittney Moore	Kirk LLC	UnitedHealthcare	36617.0455756956	

The healthcare dataset was imported into Power BI and opened in Power Query for data preparation. The dataset contains patient details, medical conditions, admission information, hospital details, insurance provider, billing amount, medication and test results.

Cleaning Categorical Data

Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
1	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal
2	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive
3	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal
4	37909.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal
5	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal
6	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal
7	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
8	45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive
9	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal
10	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive
11	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal
12	7999.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive
13	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal
14	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal
15	40701.59923	182	Urgent	13-10-2021	Paracetamol	Normal
16	12263.35743	465	Elective	11-01-2023	Aspirin	Inconclusive
17	24499.8479	114	Elective	14-07-2020	Paracetamol	Normal
18	17440.46544	449	Urgent	02-04-2020	Paracetamol	Abnormal
19	18843.02302	260	Elective	14-03-2021	Aspirin	Abnormal
20	23762.20358	465	Elective	22-11-2022	Penicillin	Normal
21	42.51458855	115	Elective	02-07-2023	Aspirin	Normal
22	17695.91162	295	Urgent	09-02-2020	Lipitor	Normal
23	5998.102908	327	Urgent	10-06-2022	Lipitor	Inconclusive
24	25250.05243	119	Urgent	12-08-2021	Lipitor	Inconclusive

The categorical text columns were cleaned using the Trim and Clean transformations. This was done to remove unnecessary spaces and unwanted characters and to make the text values more consistent for analysis.

Standardizing Names

The screenshot shows the Power Query Editor interface. The 'Name' column is selected, and the 'Capitalize Each Word' transformation is applied. The 'APPLIED STEPS' pane on the right shows the sequence of transformations: Source, Promoted Headers, Changed Type, Trimmed Text, Cleaned Text, Trimmed Text1, Cleaned Text1, Trimmed Text2, Cleaned Text2, Trimmed Text3, Cleaned Text3, Trimmed Text4, Cleaned Text4, Trimmed Text5, Cleaned Text5, and Capitalized Each Word. The data table shows the following rows:

	Name	Age	Gender	Medical Condition	Date of Admission	Doctor
1	Bobby Jackson	30	Male	Cancer	31-01-2024	Matthew Smith
2	Leslie Terry	62	Male	Obesity	20-08-2019	Samantha Davies
3	Danny Smith	76	Female	Obesity	22-09-2022	Tiffany Mitchell
4	Andrew Watts	28	Female	Diabetes	18-11-2020	Kevin Wells
5	Adrienne Bell	43	Female	Cancer	19-09-2022	Kathleen Hanna
6	Emily Johnson	36	Male	Asthma	20-12-2023	Taylor Newton
7	Edward Edwards	21	Female	Diabetes	03-11-2020	Kelly Olson
8	Christina Martinez	20	Female	Cancer	28-12-2021	Suzanne Thomas
9	Jasmine Aguilar	82	Male	Asthma	01-07-2020	Daniel Ferguson
10	Christopher Berg	58	Female	Cancer	23-05-2021	Heather Day
11	Michelle Daniels	72	Male	Cancer	19-04-2020	John Duncan
12	Aaron Martinez	38	Female	Hypertension	13-08-2023	Douglas Mayo
13	Connor Hansen	75	Female	Diabetes	12-12-2019	Kenneth Fletcher
14	Robert Bauer	68	Female	AB+	22-05-2020	Theresa Freeman
15	Brooke Brady	44	Female	AB+	08-10-2021	Roberta Stewart
16	Ms. Natalie Gamble	46	Female	AB-	01-01-2023	Maria Dougherty
17	Haley Perkins	63	Female	A+	23-06-2020	Erica Spencer
18	Mrs. Jamie Campbell	38	Male	AB-	08-03-2020	Justin Kim
19	Luke Burgess	34	Female	A-	04-03-2021	Justin Moore Jr.
20	Daniel Schmidt	63	Male	B+	15-11-2022	Denise Galloway
21	Timothy Burns	67	Female	A-	28-06-2023	Krista Smith
22	Christopher Bright	48	Male	B+	21-01-2020	Gregory Smith
23	Kathryn Stewart	58	Female	O+	12-05-2022	Vanessa Newton
24	Dr. Eileen Thompson	59	Male	A+	02-08-2021	Donna Martinez Mc
25						

The Name column contained names with inconsistent capitalization. The Capitalize Each Word transformation was applied to make the names follow a consistent format.

Creating Age Group

The screenshot shows the Power Query Editor interface. The 'Add Conditional Column' dialog box is open, allowing the user to create a new column based on existing data. The 'New column name' is set to 'Custom'. The conditions are defined as follows:

Condition	Operator	Value	Output
If Age	is less than or equal to	18	0-18
Else if Age	is less than or equal to	35	19-35
Else if Age	is less than or equal to	60	36-60
Else			60+

The background table shows patient data with columns: Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, and Doctor. The 'Age' column is being used to create the 'Age Group' column.

A conditional column named Age Group was created based on the patient's age. The age values were divided into four groups: 0-18, 19-35, 36-60 and 60+. This helps in analysing patients based on different age ranges.

Creating Dim Patient

The screenshot shows the Power Query Editor interface with the 'Dim_Patient' table. The table contains the following columns: Name, Age, Gender, Blood Type, and Age Group. The 'Age Group' column is highlighted in green. The table has 25 rows of patient data.

Name	Age	Gender	Blood Type	Age Group
Bobby Jackson	30	Male	B-	19-35
Leslie Terry	62	Male	A+	60
Danny Smith	76	Female	A-	60
Andrew Watts	28	Female	O+	19-35
Adrienne Bell	43	Female	AB+	36-60
Emily Johnson	36	Male	A+	36-60
Edward Edwards	21	Female	AB-	19-35
Christina Martinez	20	Female	A+	19-35
Jasmine Aguilar	82	Male	AB+	60
Christopher Berg	58	Female	AB-	36-60
Michelle Daniels	72	Male	O+	60
Aaron Martinez	38	Female	A-	36-60
Connor Hansen	75	Female	A+	60
Robert Bauer	68	Female	AB+	60
Brooke Brady	44	Female	AB+	36-60
Ms. Natalie Gamble	46	Female	AB-	36-60
Haley Perkins	63	Female	A+	60
Mrs. Jamie Campbell	38	Male	AB-	36-60
Luke Burgess	34	Female	A-	19-35
Daniel Schmidt	63	Male	B+	60
Timothy Burns	67	Female	A-	60
Christopher Bright	48	Male	B+	36-60
Kathryn Stewart	58	Female	O+	36-60
Dr. Eileen Thompson	59	Male	A+	36-60
Paul Henderson	72	Female	AB+	60

A separate Dim_Patient table was created by keeping the patient-related columns. The table contains Name, Age, Gender, Blood Type and Age Group. This separates patient information from the main billing data.

Adding Patient ID

The screenshot shows the Power Query Editor interface. The main area displays a table named 'Table.ReorderColumns(#Renamed Columns1',('Patient ID', 'Name', 'Age', 'Gender', 'Blood Type', 'Age Group'))'. The table has 25 rows and 10 columns. The first column is 'Patient ID', which is a new index column. The other columns are 'Name', 'Age', 'Gender', 'Blood Type', and 'Age Group'. The table is filtered to show 25 rows. The 'Patient ID' column contains values from 1 to 25. The 'Name' column contains patient names. The 'Age' column contains ages. The 'Gender' column contains 'Male' or 'Female'. The 'Blood Type' column contains blood types like 'B-', 'A+', 'AB+', etc. The 'Age Group' column contains age ranges like '19-35', '36-60', etc. The right sidebar shows the 'Query Settings' for 'Dim_Patient', including 'Properties' and 'Applied Steps'. The 'Applied Steps' list includes 'Reordered Columns1'.

Patient ID	Name	Age	Gender	Blood Type	Age Group
1	Bobby Jackson	30	Male	B-	19-35
2	Leslie Terry	62	Male	A+	60
3	Danny Smith	76	Female	A-	60
4	Andrew Watts	28	Female	O+	19-35
5	Adrienne Bell	43	Female	AB+	36-60
6	Emily Johnson	36	Male	A+	36-60
7	Edward Edwards	21	Female	AB-	19-35
8	Christina Martinez	20	Female	A+	19-35
9	Jasmine Aguilar	82	Male	AB+	60
10	Christopher Berg	58	Female	AB-	36-60
11	Michelle Daniels	72	Male	O+	60
12	Aaron Martinez	38	Female	A-	36-60
13	Connor Hansen	75	Female	A+	60
14	Robert Bauer	68	Female	AB+	60
15	Brooke Brady	44	Female	AB+	36-60
16	Ms. Natalie Gamble	46	Female	AB-	36-60
17	Hailey Perkins	63	Female	A+	60
18	Mrs. Jamie Campbell	38	Male	AB-	36-60
19	Luke Burgess	34	Female	A-	19-35
20	Daniel Schmidt	63	Male	B+	60
21	Timothy Burns	67	Female	A-	60
22	Christopher Bright	48	Male	B+	36-60
23	Kathryn Stewart	38	Female	O+	36-60
24	Dr. Eileen Thompson	59	Male	A+	36-60
25	Paul Henderson	72	Female	AB+	60

A new index column was added to the Dim_Patient table and renamed as Patient ID. This provides a unique identifier for the patient records and can be used to connect the patient dimension with the billing table.

Creating Dim Admission

The screenshot shows the Power Query Editor interface. The main area displays a table named 'Table.RemoveColumns(#Reordered Columns',('Name', 'Age', 'Gender', 'Blood Type', 'Age Group', 'Date of Admission', 'Insurance Provider'))'. The table has 25 rows and 7 columns. The columns are 'Medical Condition', 'Doctor', 'Hospital', 'Room Number', 'Admission Type', 'Medication', and 'Test Results'. The table is filtered to show 25 rows. The 'Medical Condition' column contains various conditions like 'Cancer', 'Obesity', 'Diabetes', etc. The 'Doctor' column contains doctor names. The 'Hospital' column contains hospital names. The 'Room Number' column contains room numbers. The 'Admission Type' column contains admission types like 'Urgent', 'Emergency', 'Elective', etc. The 'Medication' column contains medication names like 'Paracetamol', 'Ibuprofen', 'Aspirin', etc. The 'Test Results' column contains test results like 'Normal', 'Abnormal', 'Inconclusive', etc. The right sidebar shows the 'Query Settings' for 'Dim_Admission', including 'Properties' and 'Applied Steps'. The 'Applied Steps' list includes 'Removed Columns'.

Medical Condition	Doctor	Hospital	Room Number	Admission Type	Medication	Test Results
Cancer	Matthew Smith	Sons and Miller	328	Urgent	Paracetamol	Normal
Obesity	Samantha Davies	Kim Inc	265	Emergency	Ibuprofen	Inconclusive
Obesity	Tiffany Mitchell	Cook PLC	205	Emergency	Aspirin	Normal
Diabetes	Kevin Wells	Hernandez Rogers and Vang,	450	Elective	Ibuprofen	Abnormal
Cancer	Kathleen Hanna	White-White	458	Urgent	Penicillin	Abnormal
Asthma	Taylor Newton	Nunez-Humphrey	389	Urgent	Ibuprofen	Normal
Diabetes	Kelly Olson	Group Middleton	389	Emergency	Paracetamol	Inconclusive
Cancer	Suzanne Thomas	Powell Robinson and Valdez,	277	Emergency	Paracetamol	Inconclusive
Asthma	Daniel Ferguson	Sons Rich and	316	Elective	Aspirin	Abnormal
Cancer	Heather Day	Padilla Walker	249	Elective	Paracetamol	Inconclusive
Cancer	John Duncan	Schaefer-Porter	394	Urgent	Paracetamol	Normal
Hypertension	Douglas Mayo	Lyons Blair	288	Urgent	Lipitor	Inconclusive
Diabetes	Kenneth Fletcher	Powers Miller, and Flores	234	Emergency	Penicillin	Abnormal
Asthma	Theresa Freeman	Rivera-Gutierrez	309	Urgent	Lipitor	Normal
Cancer	Roberta Stewart	Morris-Arellano	282	Urgent	Paracetamol	Normal
Obesity	Maria Dougherty	Cline-Williams	463	Elective	Aspirin	Inconclusive
Arthritis	Erica Spencer	Cervantes-Wells	214	Elective	Paracetamol	Normal
Obesity	Justin Kim	Torres, and Harrison Jones	449	Urgent	Paracetamol	Abnormal
Hypertension	Justin Moore Jr.	Houston PLC	260	Elective	Aspirin	Abnormal
Asthma	Denise Galloway	Hammond Ltd	465	Elective	Penicillin	Normal
Asthma	Krista Smith	Jones LLC	215	Elective	Aspirin	Normal
Asthma	Gregory Smith	Williams-Davis	295	Urgent	Lipitor	Normal
Arthritis	Vanessa Newton	Clark-Mayo	327	Urgent	Lipitor	Inconclusive
Asthma	Donna Martinez MD	Sons Smith	219	Urgent	Lipitor	Inconclusive
Obesity	Matthew Smith	White-White	328	Urgent	Paracetamol	Normal

The Dim_Admission table was created to store admission-related information. The table contains Admission ID, Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number and Test Results.

Adding Admission ID

The screenshot shows a data transformation tool interface. The main table has the following columns: Hospital, Room Number, Admission Type, Medication, Test Results, and Admission ID. The Admission ID column is highlighted in green. The interface includes a menu bar (File, Home, Transform, Add Column, View, Tools, Help), a toolbar with various data manipulation icons, a query editor showing the formula 'Table.RenameColumns(#Added Index, {"Index", "Admission ID"})', and a properties panel on the right with sections for 'PROPERTIES' and 'APPLIED STEPS'.

	Hospital	Room Number	Admission Type	Medication	Test Results	Admission ID
1	Sons and Miller	328	Urgent	Paracetamol	Normal	1
2	Kim Inc	265	Emergency	Ibuprofen	Inconclusive	2
3	Cook PLC	205	Emergency	Aspirin	Normal	3
4	Hernandez Rogers and Vang,	450	Elective	Ibuprofen	Abnormal	4
5	White-White	458	Urgent	Penicillin	Abnormal	5
6	Nunez-Humphrey	389	Urgent	Ibuprofen	Normal	6
7	Group Middleton	389	Emergency	Paracetamol	Inconclusive	7
8	Powell Robinson and Valdez,	277	Emergency	Paracetamol	Inconclusive	8
9	Sons Rich and	216	Elective	Aspirin	Abnormal	9
10	Padilla Walker	249	Elective	Paracetamol	Inconclusive	10
11	Schaefer-Porter	394	Urgent	Paracetamol	Normal	11
12	Lyons-Blair	288	Urgent	Lipitor	Inconclusive	12
13	Powers Miller, and Flores	134	Emergency	Penicillin	Abnormal	13
14	Rivera-Gutierrez	309	Urgent	Lipitor	Normal	14
15	Morris-Arellano	182	Urgent	Paracetamol	Normal	15
16	Cline-Williams	465	Elective	Aspirin	Inconclusive	16
17	Cervantes-Wells	214	Elective	Paracetamol	Normal	17
18	Torres, and Harrison Jones	449	Urgent	Paracetamol	Abnormal	18
19	Houston PLC	260	Elective	Aspirin	Abnormal	19
20	Hammond Ltd	465	Elective	Penicillin	Normal	20
21	Jones LLC	115	Elective	Aspirin	Normal	21
22	Williams-Davis	295	Urgent	Lipitor	Normal	22
23	Clark-Mayo	327	Urgent	Lipitor	Inconclusive	23
24	and Sons Smith	219	Urgent	Lipitor	Inconclusive	24
25	White-White	458	Urgent	Penicillin	Abnormal	25

An index column was added to the admission table and renamed as Admission ID. This provides an identifier for the admission records and is used for connecting the admission information with the billing data.

Creating Billing Table

Patient Provider	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
1	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal
2	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive
3	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal
4	37906.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal
5	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal
6	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal
7	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
8	45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive
9	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal
10	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive
11	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal
12	7998.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive
13	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal
14	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal
15	40701.59923	182	Urgent	13-10-2021	Paracetamol	Normal
16	12265.35743	465	Elective	11-01-2023	Aspirin	Inconclusive
17	24409.8479	114	Elective	14-07-2020	Paracetamol	Normal
18	17440.46544	449	Urgent	02-04-2020	Paracetamol	Abnormal
19	18843.02302	260	Elective	14-03-2021	Aspirin	Abnormal
20	23762.20358	465	Elective	22-11-2022	Penicillin	Normal
21	42.51458855	115	Elective	02-07-2023	Aspirin	Normal
22	17695.91162	295	Urgent	09-02-2020	Lipitor	Normal
23	5998.102908	327	Urgent	20-06-2022	Lipitor	Inconclusive
24	25250.05243	119	Urgent	22-08-2021	Lipitor	Inconclusive
25						

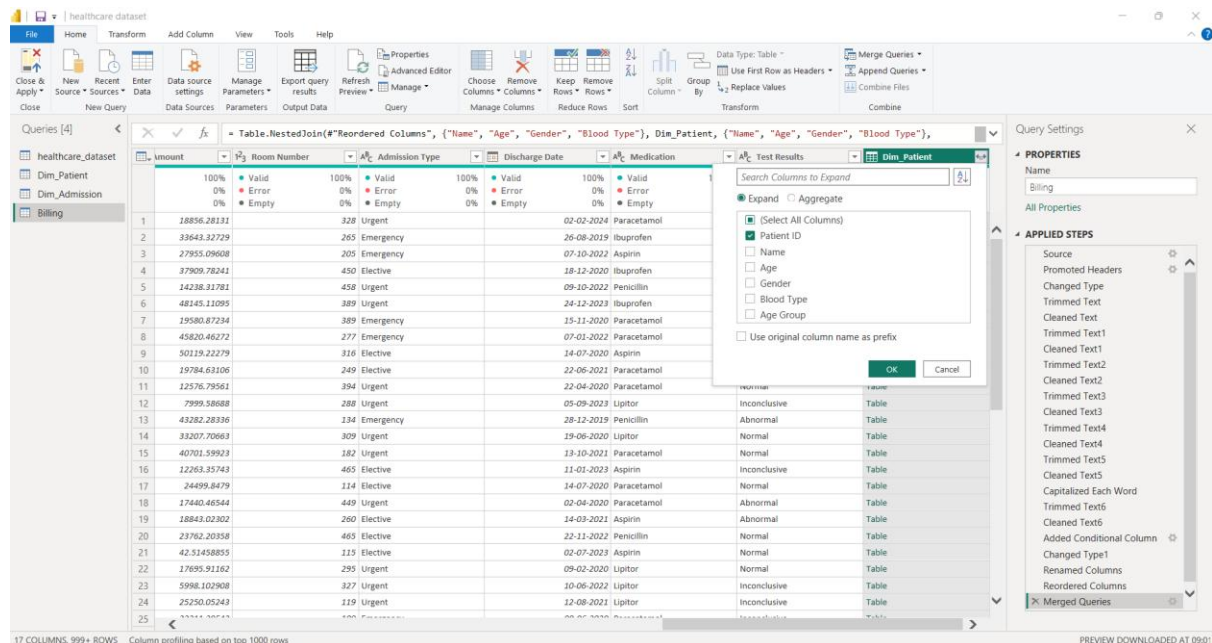
A separate Billing table was created from the prepared healthcare data. The required billing and admission-related fields were retained and the columns were organized for further modelling.

Merging with Dim Patient

Patient ID	Name	Age	Gender	Blood Type	Age Group
1	Bobby Jackson	30	Male	B-	19-35
2	Leslie Terry	62	Male	A+	60
3	Danny Smith	76	Female	A-	60
4	Andrew Watts	28	Female	O+	19-35
5	Adrienne Bell	43	Female	AB+	36-60

The Billing table was merged with the Dim_Patient table using the matching patient information. A Left Outer join was used, and all 55,500 rows from the first table were matched during the merge.

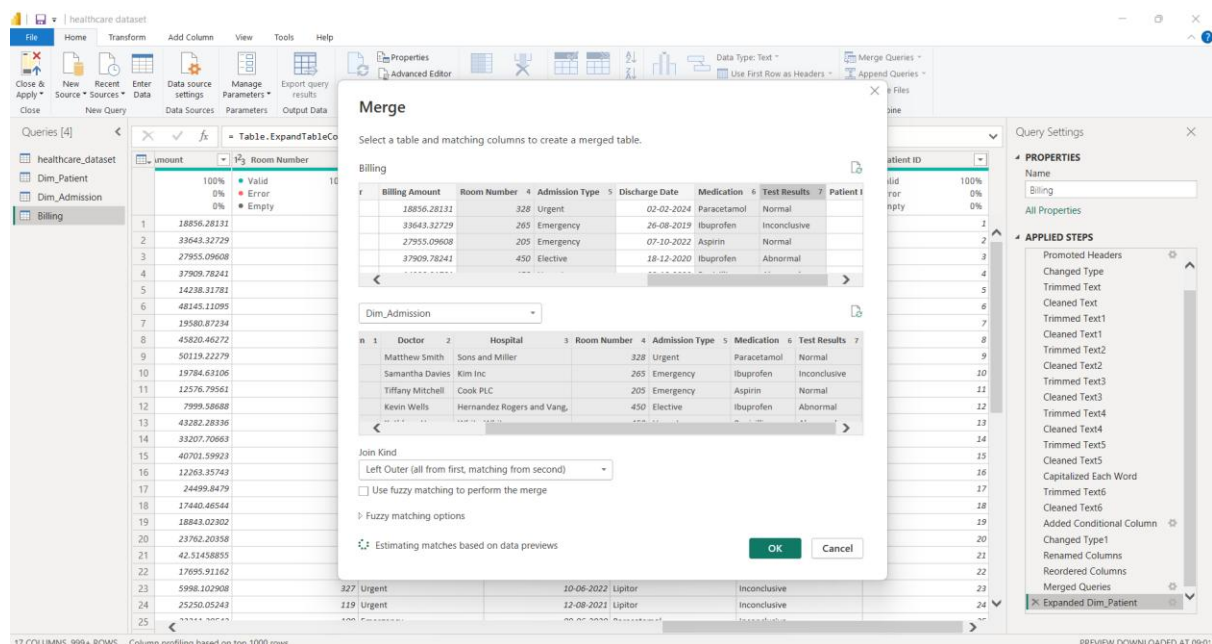
Expanding Patient ID



The screenshot shows the Power BI Desktop interface with a data table containing columns: Amount, Room Number, Admission Type, Discharge Date, Medication, Test Results, and Patient ID. The 'Patient ID' column is selected in the 'Expand' dialog box, which is open over the table. The dialog shows the 'Expand' option selected, and the 'Use original column name as prefix' checkbox is checked. The background table shows data for 25 rows, with columns: Amount, Room Number, Admission Type, Discharge Date, Medication, Test Results, and Patient ID.

After merging with Dim_Patient, the Patient ID column was selected from the merged table and expanded into the Billing table. This allows the billing records to be linked to the patient dimension.

Merging with Dim Admission



The screenshot shows the Power BI Desktop interface with a data table containing columns: Doctor, Hospital, Room Number, Admission Type, Medication, Test Results, and Patient ID. The 'Merge' dialog box is open, showing the 'Dim_Admission' table selected as the source. The 'Join Kind' is set to 'Left Outer (all from first, matching from second)'. The background table shows data for 25 rows, with columns: Doctor, Hospital, Room Number, Admission Type, Medication, Test Results, and Patient ID.

The Billing table was then merged with the Dim_Admission table using the admission information. A Left Outer join was used to bring the required admission identifier into the Billing table.

Expanding Admission ID

The screenshot displays the Power Query Editor interface. The main area shows a data table with columns: Lumber, Admission Type, Discharge Date, Medication, and Test Results. The 'Admission Type' column has a data type of 'Text' and a status of 'Valid'. The 'Discharge Date' column has a data type of 'Date' and a status of 'Valid'. The 'Medication' column has a data type of 'Text' and a status of 'Valid'. The 'Test Results' column has a data type of 'Text' and a status of 'Valid'. A dialog box titled 'Expand Columns to Expand' is open, showing a list of columns to expand. The 'Admission ID' column is selected, and the 'Use original column name as prefix' checkbox is checked. The 'APPLIED STEPS' pane on the right shows the following steps: Changed Type, Trimmed Text, Cleaned Text, Trimmed Text1, Cleaned Text1, Trimmed Text2, Cleaned Text2, Trimmed Text3, Cleaned Text3, Trimmed Text4, Cleaned Text4, Trimmed Text5, Cleaned Text5, Capitalized Each Word, Trimmed Text6, Cleaned Text6, Added Conditional Column, Changed Type1, Renamed Columns, Reordered Columns, Merged Queries, Expanded Dim_Patient, and Merged Queries1.

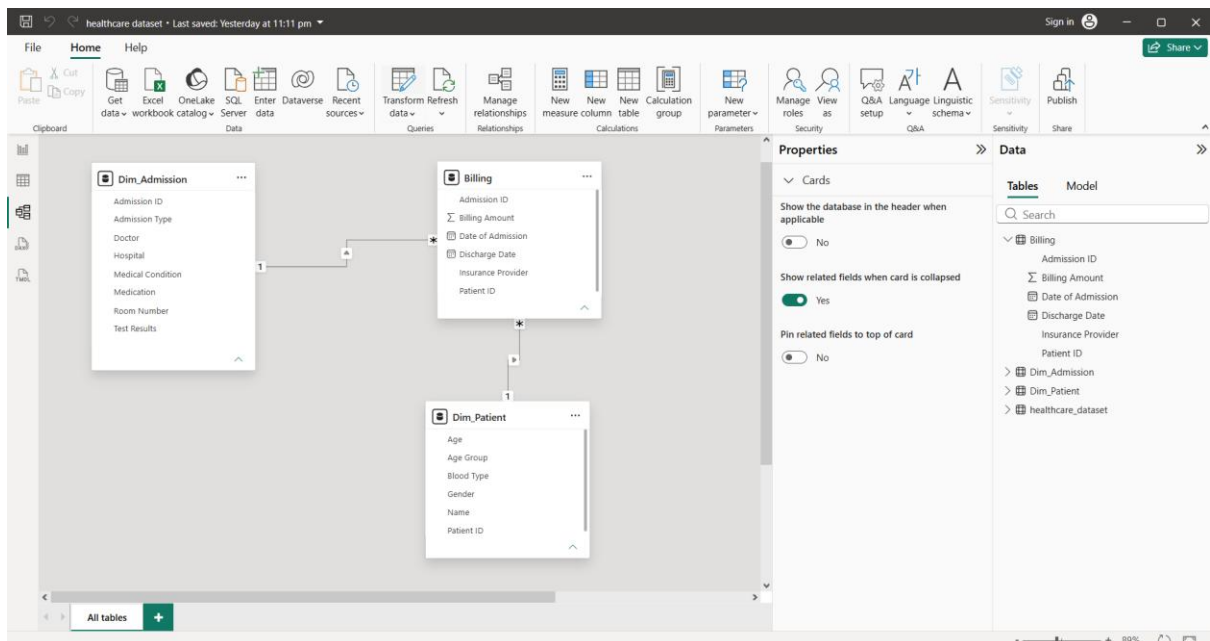
The Admission ID column from Dim_Admission was selected and expanded into the Billing table. This provides the required connection between admission details and billing records.

Final Billing Table

	Date of Admission	Insurance Provider	Billing Amount	Discharge Date	Patient ID	Admission ID
1	31-01-2024	Blue Cross	18856.28131	02-02-2024	1	1
2	20-08-2019	Medicare	33643.32729	26-08-2019	2	2
3	22-09-2022	Aetna	27955.09608	07-10-2022	3	3
4	18-11-2020	Medicare	37909.78241	18-12-2020	4	4
5	19-09-2022	Aetna	14238.31781	09-10-2022	5	5
6	19-09-2022	Aetna	14238.31781	09-10-2022	5	50145
7	19-09-2022	Aetna	14238.31781	09-10-2022	50110	5
8	19-09-2022	Aetna	14238.31781	09-10-2022	50110	50145
9	20-12-2023	UnitedHealthcare	48145.11095	24-12-2023	6	6
10	03-11-2020	Medicare	19580.87234	15-11-2020	7	7
11	28-12-2021	Cigna	45820.46272	07-01-2022	8	8
12	01-07-2020	Cigna	50119.22279	14-07-2020	9	9
13	23-05-2021	UnitedHealthcare	19784.63106	22-06-2021	10	10
14	19-04-2020	Medicare	12576.79561	22-04-2020	11	11
15	13-08-2023	Medicare	7999.58688	05-09-2023	12	12
16	12-12-2019	Cigna	43282.28336	28-12-2019	13	13
17	22-09-2020	UnitedHealthcare	33207.70663	19-08-2020	14	14
18	08-10-2021	UnitedHealthcare	40701.59923	13-10-2021	15	15
19	01-01-2023	Blue Cross	12263.35749	11-01-2023	16	16
20	23-06-2020	UnitedHealthcare	24499.8479	14-07-2020	17	17
21	08-03-2020	Cigna	17440.46544	02-04-2020	18	18
22	04-03-2021	Blue Cross	18843.02302	14-03-2021	19	19
23	15-11-2022	Cigna	23762.20358	22-11-2022	20	20
24	28-06-2023	Blue Cross	42.51458855	02-07-2023	21	21
25	21-01-2020	Aetna	17695.91162	09-02-2020	22	22

The Billing table was organized with the main fields required for analysis, including Date of Admission, Insurance Provider, Billing Amount, Discharge Date, Patient ID and Admission ID.

Final Data Model



The final data model contains Dim_Patient, Dim_Admission and Billing tables. The dimension tables are connected to the Billing table using Patient ID and

Admission ID. This structure separates patient and admission information from the main billing records and prepares the data for dashboard analysis.